# IMPACT ANALYSIS

## CR-015: Student Records Export to CSV

### 1. Change Summary

CR-015 introduces a CSV export facility to the administrator's student management interface.

Administrators will be able to export the available student records into a CSV file for administrative reporting, record-keeping and external data processing.

The exported CSV will contain the required student information, including Student ID, Registration Number, First Name, Last Name, Email and Phone Number.

The change is intended to provide a convenient way for authorized administrators to obtain student records without manually copying information from the system.

### 2. Impact Assessment

| Area | Impact | Assessment |
|---|---|---|
| Frontend | Low - Moderate | An Export CSV control and download functionality will be added to the administrator student management interface. |
| Backend | Low - Moderate | Backend student data retrieval/export support may be required depending on the existing API structure. |
| API | Low - Moderate | The existing student records API may be reused, or a dedicated export response may be introduced if required by the implementation. |
| Database | None | No database schema or existing student data structure changes are required. |
| Authentication | None | Existing authentication mechanisms remain unchanged. |
| Authorization | Low | The export functionality must remain restricted to authorized administrators. |
| Performance | Low - Moderate | CSV generation and download will process the currently available student records. |
| CRUD | Low | Existing Add, Edit and Delete student operations must continue to work normally. |
| Deployment | Low - Moderate | No database migration is expected, but frontend/backend deployment may be required depending on the final implementation. |

## 3. Frontend Impact

The main affected area is the administrator student management interface.

Potential changes include:

• Export CSV button or control.

• CSV download handling.

• User feedback when the export is successful.

• Handling of an empty student list.

• Appropriate filename for the downloaded CSV file.

• Responsive presentation of the export control.

• Preservation of existing Add, Edit and Delete functionality.

The implementation must ensure that the exported student information corresponds to the correct records displayed or retrieved from the system.

## 4. Backend, API and Database Impact

The existing student records functionality will be reviewed to determine whether the current API response can be reused for CSV generation.

If the existing student list API provides all required fields, the frontend may generate the CSV using the retrieved records.

If backend-generated export functionality is required, a suitable protected export endpoint may be introduced.

No database schema changes are expected.

No existing student records should be modified, inserted or deleted as a result of performing an export.

The export functionality must use the existing student data source and preserve the accuracy of the exported information.

## 5. Security Impact

The existing authentication and authorization mechanisms must remain unchanged.

The CSV export facility must only be accessible to authorized administrators.

The export feature must:

• Not bypass existing authentication.

• Not bypass administrator authorization.

• Not expose student information to unauthorized users.

• Export only the student information defined within the approved change scope.

• Preserve existing backend security controls.

• Prevent export functionality from being exposed through an unauthorized frontend route or direct API request.

Because the exported file contains student information, access to the export function must be treated as an administrative operation.

## 6. CRUD and Functional Impact

Existing administrator CRUD operations must continue to work correctly after

the export functionality is introduced.

Particular attention is required for:

• Adding new students.

• Editing existing students.

• Deleting students.

• Exporting the current student records.

• Ensuring exported records contain the correct student information.

• Ensuring an export operation does not modify the original student records.

The export operation must be read-only.

Downloading a CSV file must not create, update or delete student records.

## 7. Performance Impact

The expected performance impact is low to moderate for the current expected student dataset size.

CSV generation will require the system to process the student records before creating the downloadable file.

The implementation should avoid unnecessary API requests and should process the records efficiently.

For a small or moderate number of student records, the expected impact should be minimal.

If the number of student records becomes significantly larger in the future, server-side export, pagination or streaming techniques may be considered as a separate change.

## 8. Testing Impact

Manual testing should cover:

- Visibility of the Export CSV control for authorized administrators.
- Successful CSV download.
- Correct CSV filename and .csv extension.
- Correct CSV column headers.
- Student ID export.
- Registration Number export.
- First Name export.
- Last Name export.
- Email export.
- Phone Number export.
- Multiple student records.
- Accuracy of exported data compared with records displayed in the system.
- Empty student list handling.
- Opening the downloaded CSV in a spreadsheet application.
- Existing Add functionality.
- Existing Edit functionality.
- Existing Delete functionality.
- Administrator authentication.
- Administrator authorization.
- Unauthorized access to the export functionality.
- Confirmation that exporting does not modify or delete student records.

## 9. Risk Analysis

| Risk | Impact | Mitigation |
|---|---|---|
| Incorrect student data exported | High | Compare exported records against the corresponding records in the system during testing. |
| Unauthorized user accesses exported data | High | Preserve existing authentication and authorization and restrict export access to administrators. |
| CSV headers or columns are incorrect | Medium | Define and test the required CSV column structure before implementation. |
| Special characters cause incorrect CSV formatting | Medium | Apply proper CSV escaping and test names/emails containing spaces, commas or other special characters. |
| Empty student list causes export failure | Medium | Handle empty datasets safely and provide appropriate user feedback. |
| Existing CRUD functionality breaks | High | Perform regression testing for Add, Edit and Delete operations. |
| Export becomes slow for large datasets | Medium | Keep the initial scope suitable for the expected dataset size and consider server-side export for future larger datasets. |
| Export operation accidentally modifies records | High | Implement the export as a read-only operation and verify source records remain unchanged after export. |
| Responsive UI problems | Low | Test the export control on the supported screen sizes. |

## 10. Overall Impact and Recommendation

**Overall Impact: Low**

CR-015 is primarily an administrative functionality enhancement that introduces student record export to CSV.

The change is expected to have a limited impact on the existing system because no database schema changes are required and the existing authentication and authorization mechanisms can be retained.

The main implementation considerations are correct CSV formatting, accurate student data, protection of exported student information, and preservation of existing CRUD functionality.

The change is technically feasible and recommended for implementation, subject to formal approval and subsequent testing.

**Key implementation consideration:** The CSV export must contain accurate student records, must be accessible only to authorized administrators, and must not modify the original student data.